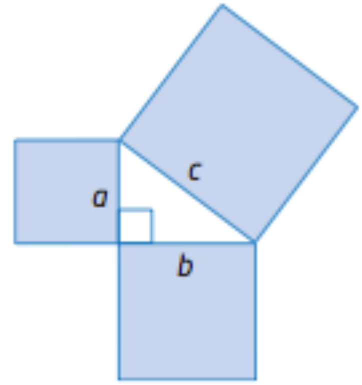
8.1 Apply the Pythagorean Theorem

MPM1D

Before we can apply the Pythagorean theorem, let's review what we know about right triangles.



DO IT NOW!!!!

A right triangle is a triangle that has one angle equal to ________ degrees.

The hypotenuse of a right triangles is:

i) The ________________________ side of a right triangle

ii) The side ________________________ the right angle

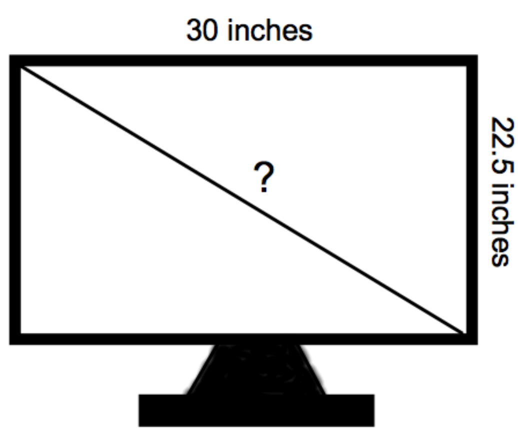
The Pythagorean theorem states: In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two shorter sides.

$$a^2 + b^2 = c^2 \quad \textbf{or} \quad c^2 = a^2 + b^2$$

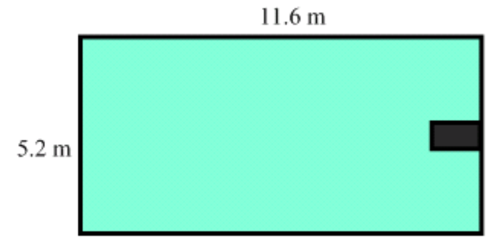
In a right triangle, we use the letters _______ and ________ to represent the two shorter sides (the legs) and the letter ________ to represent the longest side (the hypotenuse).

Part A: Use Pythagorean Theorem to Find the Hypotenuse

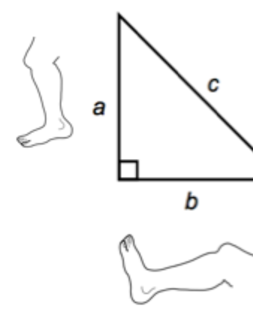
Example 1: The advertised size of a computer or television screen is actually the length of the diagonal of the screen. A television screen measures 30 inches by 22.5 inches. Determine the length of its diagonal.



Example 2: When Laurie swims laps in her rectangular pool, she decides to swim along the diagonal, so that she does not have to turn around as often. Find the distance Laurie travels by swimming once along the diagonal. Round your answer to the nearest tenth.



Part B: Use Pythagorean Theorem to Find a Leg



Example 3: Jenna is changing a light bulb. She rests a 4 meter ladder against a vertical wall so that its base is 1.4 meters from the wall. How high up the all does the top of the ladder reach? Round to the nearest tenth of a meter.

Part C: Apply the Pythagorean Theorem

Remember:

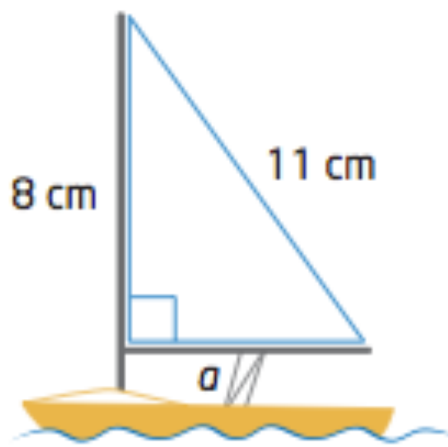
$$Area\ of\ a\ triangle = \frac{1}{2}(base)(height)$$

and

$$the\ legs\ of\ a\ right\ triangle\ are\ the\ base\ and\ height$$

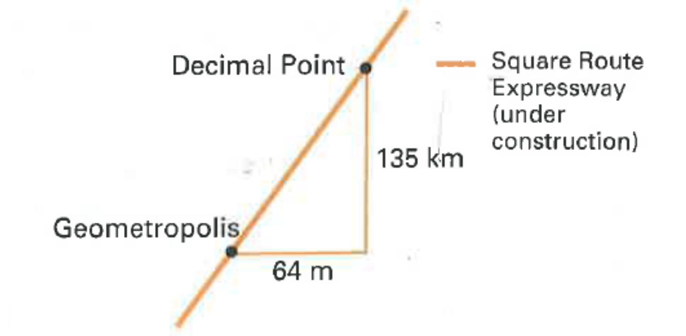
Example 4: Calculate the area of the triangular sail on the toy sailboat.

Step 1: Find the length of side a (this is the base of the triangle)



Step 2: Find the area of the sail

Example 5: Zeke drives from his house in Geometropolis to his cottage in Decimal Point. He drives 64 km east and then 135 km north. How much travel distance will he save when he drives along the new Square Rout Expressway?



Homework: Complete Worksheet